

A Project Synopsis [CSP-019]
on
JEEVANCARE - An AI Enabled Telemedicine Web
Platform

Submitted in the Partial Fulfilment of the Requirements for the award of Degree of
BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING



Submitted by
Neema Tiwari (221020101063)
Ranjeet Anand(221020101078)
Priyanshu Prakash(221020101073)
Abhishek Nayal(221020101007)

INTERNAL GUIDE:

Dr. Himanshu Verma

B.Tech -CSE

DEPARTMENT OF CSE
HARIDWAR UNIVERSITY

5TH km.Roorkee-Haridwar Canal Road, Roorkee, Uttarakhand

Batch 2022-2026

CERTIFICATE

It is to certify that the project entitled “**Jeevan Care: An AI-Enabled Telemedicine Web Platform**” which is being submitted by Neema Tiwari , under the guidance and supervision of **Dr. Himanshu Verma**, Department of Computer Science & Engineering, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech) in Computer Science & Engineering**.

The project has been completed satisfactorily during the academic year **2025–2026**, as per the rules and regulations of **Uttarakhand Technical University**. The work presented in this project is original and has not been submitted previously to any other university or institution for the award of any degree or diploma.

Dr .Himanshu Verma
Department of Computer Science and Engg.
Haridwar University, Roorkee
(Uttarakhand)

INDEX

Ch.1.Introduction

- 1.1 Intro
- 1.2 Scope Of the System
- 1.3 Existing System and problem Motivation
- 1.4 Proposed Solution
- 1.5 Advantages Of Project
- 1.6 Objectives Of Project

Ch.2.Project Category, Tools and Environment

- 2.1 Project Category
- 2.2 Tools And Technologies Used
- 2.3 Frontend , Backend and Deployment Environment
- 2.4 Hardware and Software Requirements

Ch.3.Project Development Stages

- 3.1 Feasibility study
- 3.2 System Analysis
- 3.3 Structure of Data Tables
- 3.4 System Development
- 3.5 Testing , Implementation and Maintenance

Ch.4.System Diagrams

- 4.1 Data Flow Diagrams
- 4.2 Entity-Relationship Diagram

Ch.5.Conclusion

Ch.6.Future Enhancements

Ch.7.Bibliography

INTRODUCTION

1.1 INTRODUCTION

The healthcare sector has undergone significant transformation with the rapid advancement of information technology, internet connectivity, and cloud computing. Despite these advancements, traditional healthcare systems still face numerous challenges such as overcrowded hospitals, long waiting times, limited availability of specialist doctors, and high costs associated with travel and physical consultations. These problems are more severe in rural and semi-urban areas, where access to quality healthcare facilities is limited. Patients often have to travel long distances and wait for extended periods to receive even basic medical advice, which can lead to delayed diagnosis and treatment.

Telemedicine has emerged as a promising solution to address these challenges by enabling remote consultation between patients and doctors using digital platforms. With the increasing penetration of smartphones and affordable internet services, telemedicine systems have become more practical and accessible. Telemedicine reduces the dependency on physical presence, allowing patients to receive medical guidance from the comfort of their homes. This not only saves time and cost but also helps healthcare providers manage patient loads more efficiently.

In recent years, artificial intelligence (AI) has further enhanced the potential of telemedicine by introducing intelligent features such as symptom analysis, automated triage, and decision support systems. AI can assist patients in understanding their health conditions at an early stage and help doctors make more informed decisions. The integration of AI with telemedicine platforms improves efficiency, accuracy, and user experience.

Jeevan Care is an AI-enabled telemedicine web platform developed to provide a comprehensive digital healthcare solution. The platform aims to bridge the gap between patients and healthcare professionals by offering features such as AI-based symptom checking, online appointment scheduling, and real-time video consultations. By leveraging modern web technologies, Jeevan Care ensures a user-friendly, secure, and scalable system that can be accessed through any standard web browser.

The project aligns with the vision of Digital India and e-health initiatives, focusing on improving healthcare accessibility and quality. Jeevan Care not only serves as an academic project but also demonstrates the practical application of full-stack web development, artificial intelligence, and cloud computing in solving real-world healthcare problems.

1.2 Scope of the System

The scope of the Jeevan Care system defines the functional boundaries and capabilities of the proposed telemedicine platform. The system is designed to support end-to-end telemedicine services, covering patient registration, symptom analysis, appointment scheduling, online consultation, and data management. It aims to serve as a centralized platform where patients and doctors can interact efficiently without geographical limitations.

From the patient's perspective, the system provides an easy-to-use interface for creating an account, managing personal details, and entering health-related information. One of the key features within the scope of the system is the AI-based symptom checker, which allows patients to describe their symptoms in natural language. Based on this input, the system provides preliminary guidance and helps patients decide whether they need immediate medical consultation. This feature improves awareness and reduces unnecessary hospital visits.

For doctors, Jeevan Care offers a dedicated dashboard where they can manage appointments, view patient symptom details, and conduct video consultations. Doctors can provide medical advice, suggest further tests, and maintain consultation records. This structured approach improves consultation efficiency and continuity of care.

The system also supports real-time video consultation using WebRTC technology, enabling secure and low-latency communication between patients and doctors. Medical records and appointment details are stored securely, ensuring data confidentiality and integrity.

The scope of the project can be expanded in the future to include features such as online prescription management, integration with pharmacies and diagnostic laboratories, mobile application support, and multilingual interfaces. Thus, the Jeevan Care platform is scalable and adaptable, making it suitable for both academic purposes and potential real-world deployment.

1.3 Existing System and Problem Motivation

The existing healthcare system mainly depends on traditional face-to-face consultations, where patients are required to visit hospitals or clinics for medical advice and treatment. This approach often results in

overcrowded hospitals, long waiting times, and increased workload for doctors. Patients may have to wait for several hours to receive consultation, which can be inconvenient and stressful, especially for those with minor health issues.

In rural and semi-urban areas, the problem is more severe due to the limited availability of qualified doctors and healthcare facilities. Patients often need to travel long distances to reach hospitals in urban areas, leading to additional expenses and loss of valuable time. In many cases, patients delay seeking medical attention due to these difficulties, which can worsen their health condition.

Although telemedicine systems have been introduced to reduce physical hospital visits, many existing solutions are limited in functionality. Most platforms offer only text-based chat or voice consultations, lacking real-time video interaction. The absence of video consultation reduces the quality of diagnosis, as doctors are unable to visually examine patients. Additionally, most existing systems do not include artificial intelligence support for preliminary symptom analysis.

The lack of AI assistance means that doctors must manually analyze patient symptoms during every consultation, increasing their workload and reducing efficiency. Patients also do not receive early guidance about the severity of their condition before consulting a doctor. Furthermore, issues related to data storage, security, and system scalability still exist in many healthcare platforms.

These limitations motivated the development of **Jeevan Care**, an AI-enabled telemedicine web platform designed to provide intelligent symptom analysis, real-time video consultation, and secure digital healthcare services.

1.4 Proposed System

Jeevan Care integrates a React-based frontend with a Spring Boot backend. An AI-powered chatbot assists patients in preliminary symptom analysis, while WebRTC enables real-time video consultations. The system is hosted on AWS for scalability.

1.5 Advantages of the Project

- Reduced need for physical hospital visits
- AI-assisted symptom checking
- Secure and scalable cloud deployment

- Improved accessibility for rural areas

1.6 Objectives of the Project

The primary objective of the Jeevan Care project is to design and develop a functional telemedicine web platform that enables patients and doctors to interact remotely in an efficient and user-friendly manner. The system aims to provide a reliable digital solution for healthcare consultation, reducing the dependency on physical hospital visits and improving accessibility to medical services, especially for users in remote and underserved areas.

Another important objective of the project is to integrate artificial intelligence for symptom analysis. The AI-based symptom checker allows patients to describe their health issues in natural language and receive preliminary guidance regarding possible conditions. This feature supports early awareness, helps in basic triage, and enables doctors to receive structured information before the consultation, thereby improving decision-making and reducing consultation time.

Ensuring secure and real-time video consultation is also a key objective of the project. The platform uses WebRTC technology to enable live video communication between patients and doctors through a web browser. This allows doctors to visually assess patients and provide more accurate medical advice. Data security and privacy are maintained using secure authentication mechanisms and controlled access to

CHAPTER 2 PROJECT CATEGORY, TOOLS AND ENVIRONMENT

2.1 Project Category

The Jeevan Care project falls under multiple project categories due to its use of modern technologies and its application in the healthcare domain. Primarily, it is classified as a **web-based application**, as it is developed to run on web browsers using standard web technologies. Users can access the system through the internet without the need to install any dedicated software, making the platform easily accessible on desktops, laptops, tablets, and smartphones. The web-based nature of the application ensures cross platform compatibility and ease of use.

Secondly, the project belongs to the category of **Healthcare Information Systems**. Jeevan Care manages and processes healthcare-related data such as patient information, symptoms, appointment schedules, and consultation records. It facilitates digital interaction between patients and doctors, supporting essential healthcare workflows like registration, appointment booking, and medical consultation. By digitizing these processes, the system helps improve efficiency, data organization, and continuity of care.

In addition, Jeevan Care can be classified as an **AI-enabled application** because it integrates artificial intelligence for symptom analysis. The AI-based symptom checker allows patients to enter symptoms in natural language and receive preliminary health guidance. This intelligent feature enhances user experience, supports early awareness, and assists doctors by providing structured symptom information prior to consultation.

Furthermore, the project falls under the category of **cloud-based applications**. The platform is deployed on cloud infrastructure, which provides scalability, reliability, and availability. Cloud deployment enables the system to handle multiple users simultaneously and allows future expansion without major architectural changes.

Overall, Jeevan Care is a multidisciplinary project that combines web development, artificial intelligence, cloud computing, and healthcare systems, making it highly relevant to both academic learning and real world applications.

2.2 Tools and Technologies Used

The development of the **Jeevan Care** telemedicine platform involves the use of modern and reliable tools and technologies to ensure performance, security, and scalability.

Frontend

The frontend of the application is developed using **React.js**, which is a popular JavaScript library for building user interfaces. React follows a component-based architecture, making the application modular and easy to maintain. **HTML** is used to structure the web pages, while **CSS** is used for styling and layout design. Together, these technologies provide a responsive and user-friendly interface for patients and doctors.

Backend

The backend is developed using **Spring Boot**, a Java-based framework used to build robust and scalable server-side applications. Spring Boot simplifies backend development by providing built-in configurations and support for **REST APIs**. RESTful services enable smooth communication between the frontend and backend using standard HTTP methods.

Authentication:

Firebase Authentication is used to provide secure user login and registration. It ensures proper user verification and helps protect sensitive healthcare data.

Video Communication:

WebRTC is used to enable real-time audio and video consultations directly within the web browser, ensuring low-latency and secure communication.

Deployment:

The application is deployed on **Amazon Web Services (AWS)**, which provides cloud-based scalability, reliability, and high availability.

2.3 Front-End, Back-End and Deployment Environment

The Jeevan Care telemedicine platform is developed using a well-structured full-stack architecture that separates the frontend, backend, and deployment environment to ensure scalability, maintainability, and performance.

The **front-end** of the system is responsible for user interaction and presentation. It is developed using **React.js**, which provides a responsive and dynamic user interface. React's component-based architecture allows reusable UI components such as login forms, dashboards, appointment pages, chatbot interfaces, and video consultation screens. The frontend communicates with the backend through RESTful APIs and ensures a smooth user experience across different devices and screen sizes.

The **back-end** manages the core business logic of the system. It is developed using **Spring Boot**, which handles user requests, appointment scheduling, AI symptom analysis integration, and data management. REST APIs are used to exchange data between the frontend and backend in JSON format. The backend also ensures data validation, error handling, and role-based access control for patients and doctors. The **deployment environment** uses **Amazon Web Services (AWS)** to host the application. Cloud deployment provides scalability, high availability, and reliability. AWS services allow the system to handle multiple users simultaneously and support future expansion without major infrastructure changes.

2.4 Hardware and Software Requirements

The Jeevan Care system is designed to run efficiently on standard hardware and commonly used software platforms.

Hardware Requirements:

A system with a minimum of **4 GB RAM**, a modern multi-core processor, and a stable internet connection is sufficient for development and usage. For better performance during development, **8 GB RAM** is recommended, especially when running multiple tools simultaneously.

Software Requirements:

The system can run on **Windows, Linux, or macOS** operating systems. Development requires **Node.js** for frontend dependency management, **Java Development Kit (JDK)** for running the Spring Boot backend, and an **Integrated Development Environment (IDE)** such as **VS Code or IntelliJ IDEA**. A modern web browser is required to access the application.

CHAPTER 3: PROJECT DEVELOPMENT STAGES

3.1 Feasibility Study

Before starting the development of the Jeevan Care system, a feasibility study was conducted to analyze whether the project could be successfully implemented within the given time, resources, and technical constraints. The feasibility study focuses on three main aspects: technical feasibility, economic feasibility, and operational feasibility.

From a technical feasibility perspective, the project uses well-established and widely supported technologies such as React.js, Spring Boot, Firebase, WebRTC, and AWS. These technologies are stable, welldocumented, and supported by large developer communities, which reduces technical risk. The development team possessed sufficient knowledge and learning resources to implement these technologies effectively.

From an economic feasibility point of view, the project is cost-effective. Most of the tools and frameworks used are open-source, and cloud services are utilized using free or low-cost tiers. This makes the project affordable for academic purposes without requiring expensive infrastructure.

Operational feasibility ensures that the system can be easily used by patients and doctors. The platform is designed with a simple and intuitive interface, making it easy for users to perform tasks such as booking appointments or attending consultations. Hence, the project is feasible in all aspects.

3.2 System Analysis

System analysis is a critical phase that involves understanding user requirements and defining system functionality. In Jeevan Care, requirements were gathered from two main user groups: patients and doctors.

For patients, the system should allow registration, symptom entry, appointment booking, and video consultation. For doctors, the system should support appointment management, patient information viewing, and online consultation. In addition to functional requirements, non-functional requirements such as security, performance, scalability, and usability were also considered.

Based on these requirements, the system architecture was designed. **Data Flow Diagrams (DFDs)** were used to model how data moves between users, the application, and backend services. **Entity–Relationship (ER) diagrams** were used to define the database structure and relationships between entities. This analysis ensured a clear understanding of system behavior before development began.

3.3 Structure of Data Tables

The data structure of Jeevan Care is designed to store and manage healthcare information efficiently. The primary entities in the system include **Patient**, **Doctor**, **Appointment**, and **AI Chat Session**.

The **Patient** table stores personal and basic medical information such as name, age, contact details, and login credentials. The **Doctor** table contains details such as specialization, availability, and professional information. The **Appointment** table links patients and doctors and stores appointment date, time, and status. The **AI Chat Session** table stores symptom descriptions and AI-generated responses.

This structured approach ensures data consistency, efficient retrieval, and easy scalability for future enhancements.

3.4 System Development

System development followed a modular and incremental approach. The frontend and backend were developed in parallel to save time and improve coordination. The frontend was built using React.js, focusing on responsive design and usability. The backend was developed using Spring Boot to handle business logic, API requests, and data processing.

REST APIs were implemented to enable communication between frontend and backend. WebRTC was integrated to support real-time video consultations, and Firebase Authentication was used to ensure secure user access. This modular design makes the system easy to maintain and extend.

3.5 Testing, Implementation and Maintenance

Testing was carried out at multiple levels to ensure system reliability. **Unit testing** verified individual components, while **integration testing** ensured smooth interaction between frontend and backend.

Functional testing confirmed that all features worked as expected.

After successful testing, the application was deployed on AWS. The cloud-based deployment ensures high availability and scalability. The system is designed for easy maintenance, allowing future updates, bug fixes, and feature additions without major changes.

CHAPTER 4: SYSTEM DIAGRAMS

4.1 Data Flow Diagram (DFD)

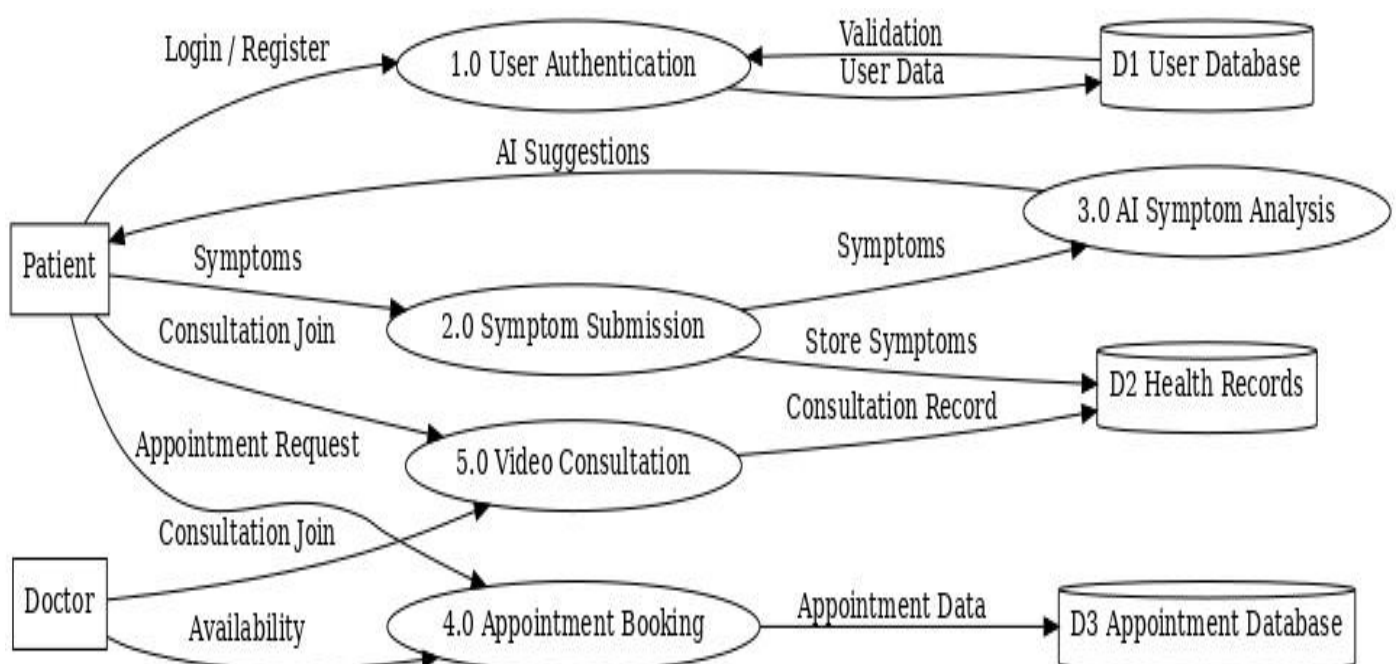
The Data Flow Diagram (DFD) illustrates how data moves through the Jeevan Care telemedicine system between users, processes, and data storage. It provides a clear picture of system functionality without focusing on implementation details.

Description of DFD

In the Jeevan Care system, the Patient and Doctor are the primary external entities. The patient interacts with the system by registering, entering symptoms, booking appointments, and attending video consultations. The doctor accesses the system to view appointments, patient details, and conduct consultations.

The main processes involved include:

- User Authentication: Verifies patient and doctor login credentials.
- Symptom Submission & AI Analysis: Patient symptoms are submitted and analyzed by the AI module.
- Appointment Booking: Patients book appointments, and doctors receive schedules.
- Consultation Handling: Real-time video consultation using WebRTC.
- Data Storage: User details, health records, and appointments are stored securely in the database. The DFD ensures smooth data movement and helps identify how information flows securely within the system.



4.2 Entity–Relationship (ER) Diagram

The Entity–Relationship (ER) Diagram represents the database structure of the Jeevan Care system. It shows entities, their attributes, and relationships between them.

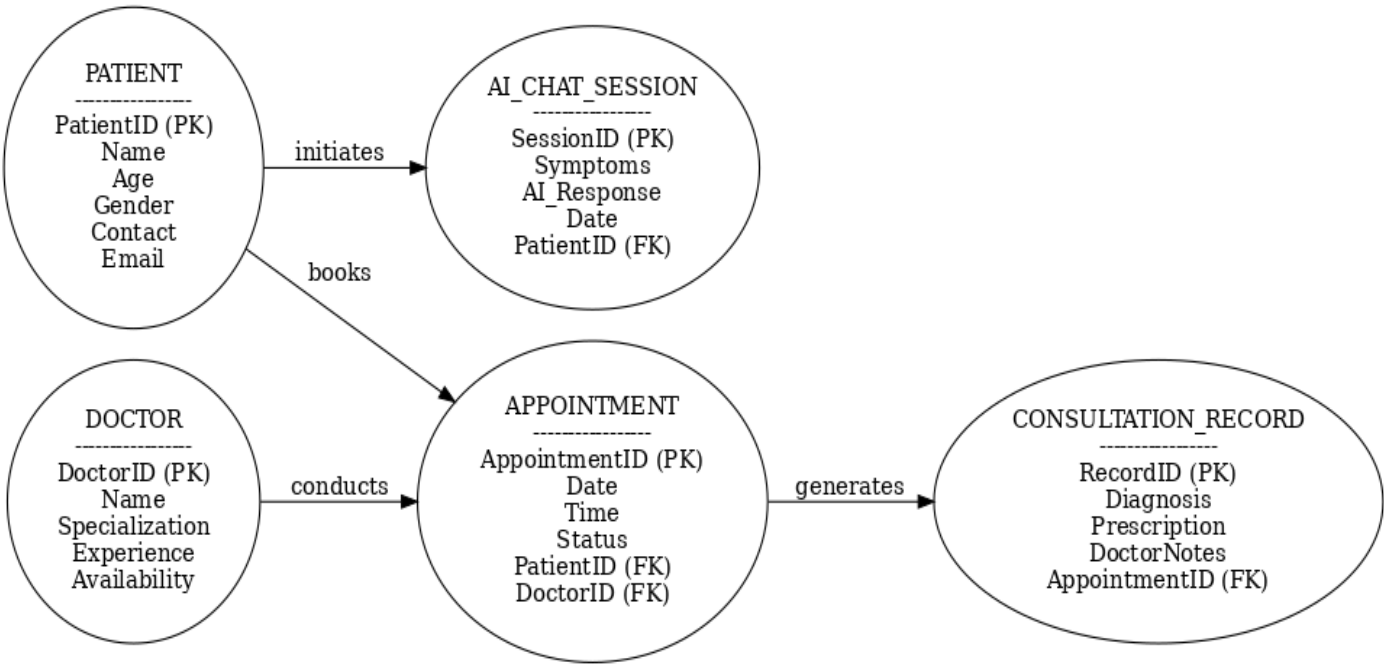
Description of ER Diagram

The main entities in the system are:

- Patient: Stores patient details such as Patient ID, Name, Age, Contact Information.
- Doctor: Stores doctor details such as Doctor ID, Name, Specialization, Availability.
- Appointment: Links Patient and Doctor, storing Date, Time, and Status.
- AI_Chat_Session: Stores symptom input and AI-generated responses.
- Consultation_Record: Stores diagnosis, prescription, and doctor notes.

A patient can book multiple appointments, and a doctor can handle multiple appointments. Each appointment is associated with one consultation record.

The ER diagram helps in database normalization, avoids redundancy, and ensures data integrity.



CHAPTER 5: CONCLUSION

The **Jeevan Care** project successfully demonstrates the design and development of an **AI-enabled telemedicine web platform** using modern web and cloud technologies. The system effectively addresses major challenges in traditional healthcare, such as overcrowded hospitals, limited access to doctors, and long waiting times.

By integrating **AI-based symptom analysis**, the platform provides patients with preliminary medical guidance before consultation. The inclusion of **real-time video consultation using WebRTC** enhances the quality of interaction between patients and doctors, enabling better diagnosis and treatment. Secure authentication and cloud deployment ensure data privacy, scalability, and system reliability.

From an academic perspective, the project provided hands-on experience in full-stack development, system design, database modeling, testing, and cloud deployment. The project meets all its objectives and serves as a strong foundation for real-world telemedicine applications.

Overall, Jeevan Care proves that digital healthcare solutions can significantly improve accessibility, efficiency, and quality of healthcare services.

CHAPTER 6: FUTURE ENHANCEMENT

Although Jeevan Care offers essential telemedicine features, there is significant scope for future improvement. Advanced **machine learning models** can be integrated to provide more accurate disease prediction and personalized health recommendations.

A dedicated **mobile application** for Android and iOS can be developed to improve accessibility and user experience. Integration with **pharmacies and diagnostic laboratories** can allow online medicine delivery and digital test reports.

Additional enhancements may include **multilingual support, voice-based chatbots**, wearable device integration, and analytics dashboards for doctors and hospitals. These features can transform Jeevan Care into a complete digital healthcare ecosystem.

REFERENCES / BIBLIOGRAPHY

1. React Team, *ReactJS Official Documentation*, Available at: <https://react.dev>
2. Pivotal Software, *Spring Boot Reference Documentation*, Available at: <https://spring.io/projects/spring-boot>
3. Google Firebase, *Firebase Authentication Documentation*, Available at: <https://firebase.google.com/docs/auth>
4. Google Firebase, *Cloud Firestore Documentation*, Available at: <https://firebase.google.com/docs/firestore>
5. W3C and WebRTC Working Group, *WebRTC Official Documentation*, Available at: <https://webrtc.org>
6. Amazon Web Services, *AWS Cloud Computing Documentation*, Available at: <https://docs.aws.amazon.com>
7. World Health Organization (WHO), *Telemedicine: Opportunities and Developments in Member States*, WHO Press, Geneva.
8. K. K. Aggarwal and Yogesh Singh, *Software Engineering*, New Age International Publishers.
9. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education.
10. Pressman, R. S., *Software Engineering: A Practitioner's Approach*, McGraw-Hill Education.
11. IEEE, *IEEE Standards for Software Engineering*, IEEE Computer Society.
12. Research articles and journals on **Telemedicine Systems, Artificial Intelligence in Healthcare, and Cloud-Based Web Applications**.